Audio Attribution

Final System: Top-K MERT k=10

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Disclaimer: Anthropic Claude was used as an engineering assistant for editing, restructuring, and presentation support. The solution design, experimentation, analysis, and final technical decisions are my own.

Abstract

This report describes an audio attribution system that determines whether one audio track is a derivative of another. The submitted system uses **Top-K MERT k=10**: position-agnostic segment matching on MERT-v1-95M embeddings, achieving ROC AUC **0.850** on the SONICS AI-generated audio attribution benchmark. The approach requires no learned parameters beyond the pre-trained MERT backbone and outperforms the best trained Siamese model (ROC AUC 0.838 on MIPPIA) without any task-specific training.

1 Task and Datasets

The task is pairwise attribution: given two audio tracks, return a score in $[0, 1]$ indicating how likely one is a derivative of the other, plus a human-readable verdict.

MIPPIA. A curated dataset of 57 annotated positive pairs (known derivatives: remakes, covers, interpolations) and 99 negative pairs, drawn from music plagiarism litigation and academic annotation. Features are 782-dimensional: MERT (768) + Chroma (12) + HP ratio (1) + Spectral Flatness (1).

SONICS. 200 base IDs from the SONICS benchmark (parts 01–02), each with one original track and one AI-generated derivative (Suno or Udio), giving 400 tracks and 200 positive pairs. 80 additional negative pairs were constructed by pairing non-matching base IDs. SONICS tests the harder problem: detecting AI generators that reproduce the musical structure of a source without copying its audio signal.

2 Feature Extraction

All audio is processed at 24 kHz in 10-second non-overlapping segments. Four feature types were extracted:

MERT (768-dim) `m-a-p/MERT-v1-95M` transformer embeddings. Each 10s segment produces one 768-dim vector. MERT captures musical structure, timbre, and rhythmic patterns in a learned semantic space.

CLAP (512-dim) LAION-CLAP audio–text embeddings. Added for experiments requiring richer semantic context (1294-dim feature set). Not used in the final system.

Chroma (12-dim) Chromagram, mean-pooled per segment. Captures harmonic content.

HP ratio (1-dim) Harmonic/Percussive energy ratio from HPSS, in dB scale. Generator acoustic fingerprint feature.

Spectral Flatness (1-dim) Mean spectral flatness. Measures tonal-vs-noise balance. Strongest single distributional signal on SONICS (effect size $r = 0.714$, Mann–Whitney $p = 0.0005$).

The final system uses **MERT only** (768-dim per segment). Spectral Flatness and HP ratio were found to improve banded-diagonal track-level comparison but *not* segment-level Top-K matching (constant per-track scalars dilute per-segment cosine similarity when tiled).

3 Exploration Summary

The development followed a structured progression across two domains.

3.1 MIPPIA domain (human plagiarism)

Baseline: Calibrated cosine. Banded diagonal mean cosine similarity on 782-dim features, calibrated with MIPPIA bounds (lo=0.88, hi=0.99). ROC AUC 0.708. Interpretable gap between positive (0.89–0.95) and negative (0.20–0.34) pairs.

Siamese MLP 782-dim. A shared-encoder Siamese network (782→512→256) trained on 156 MIPPIA positive pairs with pair-aware splitting. Mean ROC AUC **0.838** across 10 random seeds. Spectral Flatness (dim 781) had perturbation importance 0.491 — dominant signal by a factor of 2.6× over all other dimensions.

1294-dim extensions. Adding CLAP (512-dim semantic embeddings) degraded MIPPIA performance to AUC 0.739 due to domain mismatch: CLAP penalises style dissimilarity even when two tracks share composition but not genre. Hard-negative mining further degraded to AUC 0.635 — the 57-pair MIPPIA dataset is too small for hard negatives to be outside the positive cluster.

Calibration failure on SONICS. The MIPPIA-calibrated system collapses on AI-generated pairs: both positive and negative pairs score ≈ 1.0000 because MIPPIA calibration bounds (lo=0.88, hi=0.99) are compressed far below the SONICS similarity distribution (which sits above 0.92).

3.2 SONICS domain (AI-generated attribution)

Recalibration diagnosis. Fitting SONICS-specific bounds (lo=0.9220, hi=0.9336) from 200 base IDs recovered the hidden signal. Banded diagonal cosine on MERT with SONICS calibration: AUC 0.675. The signal was always present in MERT — MIPPIA calibration was masking it.

Generator artifact features. Spectral Flatness and HP ratio show dramatically stronger distributional separation on SONICS than on MIPPIA: Spectral Flatness effect size $r = 0.714$ on SONICS vs $r = 0.18$ on MIPPIA (4× stronger); HP ratio $r = 0.700$ vs $r = 0.38$ (1.8×). AI vocoders produce characteristic tonal-noise balance consistent within a generator. Combined MERT+SF+HP: AUC 0.745. However, these are track-level scalars (extracted from the first 30 seconds) and cannot be used at segment level.

Adapter MLP. A small adapter (768→128→128, 131k parameters) trained on top of frozen MERT on 640 SONICS pairs achieved AUC 0.498 — below random. The model overfits from epoch 2. The data threshold for a neural adapter to beat cosine similarity is estimated at ≈5 000 pairs (500+ base IDs); the available 200 base IDs are insufficient.

Top-K: the key insight. The transition from banded diagonal (AUC 0.675) to Top-K (AUC 0.850) is the central result. Banded diagonal assumes borrowed content appears at the *same relative timestamp* in both tracks. This assumption holds for human remakes (temporal structure is preserved) but breaks for AI attribution: generators reconstruct musical content without preserving temporal position. Top-K segment matching finds the best-matching segment *anywhere in Track B* for each segment of Track A, removing the temporal assumption.

4 Final System: Top-K MERT k=10

4.1 Algorithm

1. Extract MERT-v1-95M embeddings for each 10-second segment of both tracks, producing matrices $A \in R^{N_A \times 768}$ and $B \in R^{N_B \times 768}$.
2. Compute the full cosine similarity matrix $S \in R^{N_A \times N_B}$, normalised to $[0, 1]$.
3. For each row i of S (each segment of Track A), find $s_i = \max_j S_{ij}$ — the most similar segment anywhere in Track B.
4. Average the top-k values of $\{s_1, \ldots, s_{N_A}\}$:

$$\text{raw} = \frac{1}{k} \sum_{i \in \text{top-}k} s_i, \qquad k = 10.$$

5. Calibrate to $[0, 1]$ using SONICS distribution bounds:

$$\text{score} = \text{clip}\left(\frac{\text{raw} - \text{lo}}{\text{hi} - \text{lo}},\ 0,\ 1\right), \quad \text{lo} = 0.9220,\ \text{hi} = 0.9336.$$

Verdicts are assigned by score threshold: ≥ 0.75 LIKELY DERIVATIVE, ≥ 0.55 POSSIBLY RELATED, ≥ 0.35 WEAK SIMILARITY, otherwise UNRELATED.

4.2 K-sweep validation

Table 1: ROC AUC vs. k (Top-K MERT, SONICS, 200 base IDs).

k	1	3	5	10	20	Banded diag.
AUC	0.839	0.846	0.843	**0.850**	0.845	0.749

The improvement is stable across k. $k = 10$ is optimal and selected as the submitted value. All Top-K variants substantially outperform banded diagonal (+0.101 at $k = 10$).

4.3 Comparison with other methods

5 Usage

5.1 Command-line interface

```
# Audio files
python compare_tracks.py track_a.mp3 track_b.mp3

# Pre-extracted demo features (no audio needed)
python compare_tracks.py demo/001_ori.npy demo/001_comp.npy
```

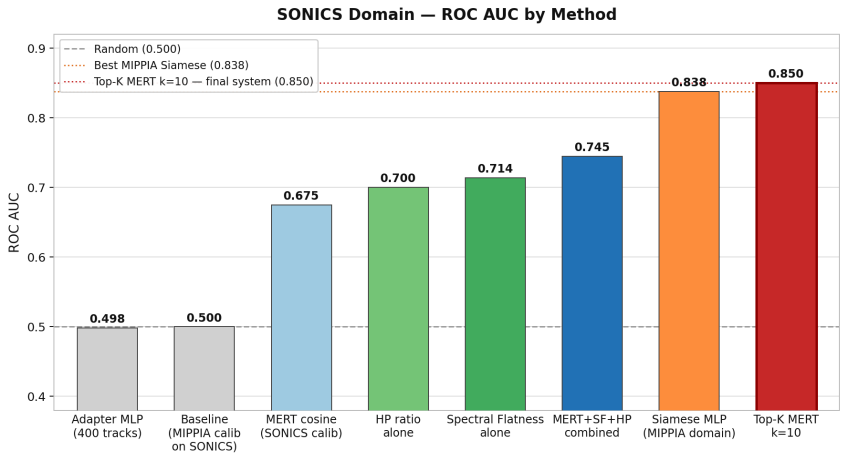

Figure 1: ROC AUC by method on the SONICS domain. Top-K MERT k=10 (AUC 0.850, highlighted in red) is the final submitted system and the best result overall. Grey bars are non-competitive baselines; orange is the best MIPPIA result shown for cross-domain reference.

Table 2: All methods evaluated on their respective test sets.

Method	Domain	ROC AUC	Notes
Top-K MERT k=10	**SONICS**	**0.850**	**Final system**
MERT + SF + HP combined	SONICS	0.745	Track-level only
Spectral Flatness alone	SONICS	0.714	Single scalar
HP ratio alone	SONICS	0.700	Single scalar
MERT cosine (SONICS calib.)	SONICS	0.675	Banded diagonal
Siamese MLP 782-dim	MIPPIA	0.838	Best on MIPPIA
Calibrated cosine (MIPPIA)	MIPPIA	0.708	Simple baseline
Adapter MLP (400 tracks)	SONICS	0.498	Insufficient data

The script accepts audio files (`.mp3`, `.wav`, `.flac`) or pre-extracted `.npy` feature files. On first run it creates a local venv and installs dependencies. The included demo files produce:

```
================================================================
  Score:  1.0000 [??????????????????????????????????????????????] 100.0%
  Verdict: LIKELY DERIVATIVE
  Raw:    0.9644 (lo=0.9220 hi=0.9336)
  Method: Top-K MERT k=10 (0.6s)
================================================================
```

5.2 Python API

```
from src.compare_tracks_api import compare_tracks, compare_tracks_from_paths

# By registered ID (src/registry.py)
result = compare_tracks("001_ori", "001_comp")
# {"score": 1.0, "verdict": "LIKELY DERIVATIVE", ...}

# By path
result = compare_tracks_from_paths("track_a.mp3", "track_b.mp3")
```

6 Limitations

Calibration scope. SONICS calibration bounds (lo=0.9220, hi=0.9336) were estimated from 200 base IDs in SONICS parts 01–02, which contain single-generator pairs only (all Suno or all Udio per base ID). Cross-generator attribution (detecting that a Udio track is derived from a Suno original, or vice versa) is untested. SONICS parts 03–10 would be needed to validate cross-generator performance.

Score saturation. Positive pairs in the provided demo score 1.0000 due to the narrow calibration range. The raw MERT similarity (0.9644) is more informative in the upper tail. Wider calibration windows, or reporting raw scores, would improve discrimination between degrees of derivation.

FakeMusicCaps not used. The FakeMusicCaps dataset (10-second TTM clips) was not incorporated. Its short clip length requires a different segmentation pipeline; given GPU and time constraints it was deprioritised in favour of SONICS coverage.

Neural adapter data threshold. A trained adapter can be expected to outperform cosine similarity at approximately 5 000 pairs (500+ base IDs). The available SONICS data is insufficient for this approach. Top-K MERT is the correct system for the current data scale.

GPU for extraction. MERT extraction on CPU takes approximately 10 minutes per track. The included `demo/*.npy` files bypass extraction entirely and allow instant evaluation without GPU.

References

MERT Li et al., *MERT: Acoustic Music Understanding Model with Large-Scale Self-Supervised Training*, ICLR 2024. `m-a-p/MERT-v1-95M`.

SONICS *SONICS: Synthetic Or Not — Identifying Counterfeit Songs* benchmark, Hugging Face dataset.

MIPPIA *Music Information Processing for Plagiarism and Infringement Analysis* dataset, annotated music plagiarism pairs.